RigPro Job Safety Assessment (JSA) Application

(Power Apps + SharePoint + OneDrive/SharePoint Storage)

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1) Platform Architecture

The application must be built using:

Power Apps (Canvas App)

For all screens, logic, user input, business rules.

SharePoint Lists

Used for:

1. **Job Master List** (synced from Salesforce) – COULD BE MANUAL IF DIFFICULT
    - Contains Job Name, Job Number, Service Log Number, Boat Name
2. **JSA Records List**
    - Stores metadata for each created JSA + JSON of selections
3. **PDF storage (SharePoint Folder)**
    - `/<Service Log Number>/H&S/`
    - The app will save final JSA PDFs here

Power Automate

Used for:

- PDF generation (using HTML → PDF)
- Saving PDF to `/Jobs/<ServiceLog>/H&S/`
- Writing final JSON to JSA Records list

Data Files

Loaded into Power Apps as static data sources:

- **Risk_Table** (hazards + matrix + PPE)
- **Questionnaire** (questionnaire + hazard triggers)
- **JSA template** (HTML layout derived from PDF)

OR SHOULD THESE BE IN A SHAREPOINT LIST WE CAN EDIT?

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2) User Roles

Technician / Rigger

- Create JSAs
- Enter job details via dropdown and manual entry
- Enter steps
- Complete questionnaire
- Review suggested hazards, controls, PPE
- Approve and generate PDF

Supervisor / PM

- Review/edit risk matrix
- Approve final JSA
- Sign PDF

Admin

- Manage data sources (hazards, questionnaire)
- Manage SharePoint connectors
- Maintain templates

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3) Data Sources

3.1 Job Master List (SharePoint)

Pulled from Salesforce nightly (existing connector or Flow)

Columns:

- JobName (Text)
- JobNumber (Text)
- ServiceLogNumber (Text)
- Boat Name (Text)

3.2 Risk_Table

Source: **Risk_Table.xlsx**
Contains:

- Hazard ID
- Hazard
- JSA Risk Level
- Control Measures
- Recommended PPE
- Risk matrix default fields (pre/post L/S/Score)

3.3 questionnaire

Source: **questionnaire.xlsx**
Contains:

- Question ID
- Question
- AnswerType
- Hazard IDs triggered
- Notes/Logic

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4) Power Apps — Screen-by-Screen Build

4.1 Screen: Home Page

- Buttons: **Create New JSA** → `Navigate(scr_JobDetails)`; **Use a Template** → `Navigate(scr_SelectTemplate)`.
- Short blurb: "Start a new JSA or apply a saved template."

4.2 Screen: Template Selection

- Gallery items: `JSA_Templates` library (TemplateName, LastModified).
- **Use Template**: loads `JSONData` into app collections; then `Navigate(scr_JobDetails)`.

4.3 Screen: Job Details

Controls:

- **Dropdown: Select Job**
  - `Items = JobMasterList`
  - Display job name
  - OnSelect: Patch job metadata fields (Job Number, Boat Name, Service Log Number)
- **Text fields auto-populated:**
  - Job Number
  - Boat Name
  - Service Log Number
- **Text fields Manually Populated:**
  - Date
  - Location
- **Technician/Role Repeater**
  - Collection with Add/Remove rows: Collect(technicians, {Name:"", Role:""})

Validation:

- Job must be selected
- At least one technician entered

Button → Next (Steps screen)

4.4 Screen: Steps Entry

UI:

- Gallery listing steps
- "Add Step" button → `Collect(steps, {StepText:""})`
- Text Input inside gallery to edit step text
- Delete icon

Button → Next (Questionnaire)

4.5 Screen: Questionnaire

UI:

Gallery pulling from `questionnaire`

Each row:

- Question text
- Yes/No toggle
- Store answers in `colAnswers`

Logic:

Store hazard triggers:

```
If(Toggle.Value = true,
   Collect(colHazards, Split(ThisItem.'Hazard IDs triggered', ";"))
)
```

4.6 Screen: Analysis (AI/Logic)

Functions used:

1. Hazards triggered by questionnaire

```
colSelectedHazards = Distinct(colHazards, Value)
```

2. Hazards triggered by steps (keyword inference)

Example Patterns:

- Dust: `"sand","dust","cut","grind"` → Haz 22
- Resins: `"resin","epoxy","solvent","acetone"` → Haz 23
- Generator: `"generator"` → Haz 24
- Oven: `"oven","hot box","cure"` → Haz 25
- Crane keywords → Haz 21
- Height cues → Haz 18/19 depending on Q12/Q13
- Ladder → Haz 8 or Haz 19

Implementation:

```
ForAll(colSteps,
   If(Find("sand", StepText) > 0 Or Find("dust", StepText) > 0,
      Collect(colHazards, "22")
   )
)
```

Repeat for all keywords.

3. Combine hazards

```
colAllHazards = Distinct(colHazards, Value)
```

4. Lookup full hazard data

```
colHazardObjects = AddColumns(
   colAllHazards,
   "HazardRec", LookUp(Risk_Table, 'Hazard No' = Value)
)
```

5. Copy controls and PPE into a review structure

```
Collect(colReview,
   {
     HazardId: HazardRec.'Hazard No',
     HazardName: HazardRec.Hazard,
     Controls: HazardRec.'Control Measures',
     PPE: HazardRec.'Recommended PPE',
     PreL: HazardRec.'Pre Likelihood (1-3)',
     PreS: HazardRec.'Pre Severity (1-3)',
     PostL: "",
     PostS: ""
   }
)
```

Output saved:

- `colReview` (hazards + matrix)
- `colPPE` (unique list from PPE fields)

Next → Review screen

4.7 Screen: Review & Edit

Editable fields:

- Controls (text input or multi-line)
- PreL / PreS
- PostL / PostS
- PPE checkboxes (collected from all selected hazards)

Auto-calc:

```
PreScore = PreL * PreS
PostScore = PostL * PostS
NewRiskLevel = If(PostScore <=2, "Low", If(PostScore <=4, "Moderate",
"High"))
```

Allow manual adding missing risks/hazards/PPE

Button → Accept & Generate PDF

4.8 PDF Generation (Power Automate Flow)

Trigger: Power Apps button
Inputs:

- Job details
- Steps
- Hazards (colReview)
- PPE list
- Technician list

Flow Actions:

1. **Compose HTML** using layout matching **JSA template** (PDF you provided)
2. **Convert HTML to PDF** (OneDrive or Premium connector)
3. **Save PDF to SharePoint:**

Path:
```
/<ServiceLogNumber>/H&S/
```
Filename:
```
JSA_<JobNumber>_<BoatName>_<<Date>.pdf
```

4. **Write JSON + metadata** to JSA Records list.
5. Return PDF URL to Power Apps.

Button → Save as Template

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5) Data Storage Model (SharePoint)

JSA Records List

Columns:

- Title (Job Number)
- JobName
- ServiceLogNumber
- CreatedBy
- CreatedDate
- JSONData (multiline text)
- PDFLink (hyperlink)

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6) Risk Matrix Logic (Same as earlier)

Pre-control seeding (based on JSA level)

- High → 3×3
- Medium → 2×2
- Low → 1×2

Post-control suggestions:

- Reduce Likelihood by 1 level where reasonable
- Severity stays mostly constant

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7) PPE Aggregation

Take **all PPE entries** from triggered hazards in **Risk_Table** and dedupe into a final checklist.

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8) QA / Acceptance Checklist

1. Power Apps loads Job list from SharePoint.
2. Job details auto-populate from Salesforce including Service Log Number.
3. Review screen editable and risk matrix recalculates.
4. PDF matches JSA template exactly.
5. PDF saved to:
   `/Jobs/<ServiceLog>/H&S/`JSA JSON stored successfully.

6. JSA templates can be used and saved.

Risk Table

Hazard No	Hazard	JSA Risk Level	Control Measures	Recommended PPE
1	Manual Handling	Low	Correct lifting techniques; Team lift >20kg or awkward; Gloves; Safety footwear	Gloves; Safety footwear
2	Hand Tools	Low	Safe operation; Safe storage; Correct PPE; Do not use damaged tools	Gloves; Eye protection; Safety footwear
3	Trip Hazards	Low	Good housekeeping; Safe stacking/storage; Remove rubbish; Safety footwear	Safety footwear
4	Pinch and Nip Hazards	Medium	Competent assembly; Trained removal; Gloves	Gloves; Safety footwear
5	Lifting Equipment	High	Certified slings/shackles; No damage	Helmet; Gloves; Eye protection; Safety footwear
6	Feet Hazards	Low	Awareness of hazards; Enclosed footwear; Avoid under-height work	Safety footwear
7	Slip Hazards	Medium	Awareness; Tidy area; Wipe spills; Remove rubbish; Footwear/overshoes; Lifelines	Safety footwear; Overshoes
8	Ladder	Medium	Trained users; Correct type; Certified; Correct positioning; Isolate below	Safety footwear; Helmet
9	Forklift	High	Trained operators; Approved attachments; Signage	Hi-Vis; Safety footwear
10	Hydraulics	Medium	Report issues; Trained workers	Gloves; Safety footwear
11	Traffic	Low	Cordon with cones; Hi-Vis	Hi-Vis; Safety footwear

12	Open Hatches	Medium	Advise personnel; Close hatches; Trained only	Safety footwear; Helmet
13	Weather/Environment	Medium	Sun protection; Lightning stop; Dust mask	Sun protection; Dust mask; Safety footwear
14	Items Dropped from Height	High	Lanyards; Clear below; Helmets/footwear	Helmet; Safety footwear
15	High Load Equipment	Medium	Trained only; Assess area; Notify	Gloves; Safety footwear
16	Winches	Medium	Trained operators; No loose clothing	No loose clothing; Safety footwear; Gloves
17	Pedestrians	Low	Load/unload away; Cordon; Escort pedestrians; Hi-Vis	Hi-Vis; Safety footwear
18	Working at Height (>2m)	High	Trained only; Working Aloft procedures; Helmet/harness; Radios; Suspension trauma plan	Helmet; Harness + Lanyard; Radios; Safety footwear
19	Working at Height (<2m)	Medium	Trained only; Edge awareness; 3 points of contact; Short ladder work	Safety footwear; Helmet
20	Electricity	Medium	Use near power; RCDs; Keep dry; Report damage	Gloves; Safety footwear; RCD device
21	Cranes	High	Qualified operator; Certified crane; Outriggers; Clear slew area; Hook latch; Taglines; No one under load	Helmet; Gloves; Safety footwear; Hi-Vis
22	Air particulates (dust)	Medium	Extraction/ventilation; Enclose area; Respiratory+eye protection	Respirator; Eye protection; Gloves; Safety footwear
23	Resins/Solvents	Medium	Ventilation; PPE; Eyewash; Safe containers	Gloves; Respirator; Eye protection; Safety footwear

24	Carbon monoxide (generators)	High	Use outdoors; Avoid enclosed spaces; Airflow control	Respirator; Safety footwear
25	Fire from ovens (curing)	High	Temperature control; Keep combustibles away; Heat PPE	Heat-resistant gloves; Eye protection; Safety footwear

Questionaire

Question ID	Question	Hazard IDs Triggered
1	Is manual handling of heavy, large or awkward items required?	1
2	Will hand tools be used?	2
3	Are there housekeeping or trip hazards in the work area?	3
4	Will certified lifting equipment be used?	5
5	Are open hatches/locker lids/deck hardware present?	6;12
6	Are there slip hazards?	7
7	Will ladders be used?	8
8	Will a forklift be operated or nearby?	9
9	Will hydraulic equipment be operated or serviced?	10
10	Is there vehicle/yard traffic interfacing with the area?	11;17
11	Are environmental factors present (sun, lightning, wet, dust)?	13
12	Will any work be carried out at height greater than 2 metres?	18;14
13	Will any work be carried out at height below 2 metres?	19;14
14	Is there a risk of items being dropped from height?	14
15	Will equipment be operated under high load?	15
16	Will winches be operated?	16
17	Will pedestrians or public be present nearby?	17
18	Will electrical equipment or temporary power be used?	20
19	Will a crane be used for lifting?	21;14
20	Will taglines be required to control load swing?	21;5
21	Will slings/shackles/chain blocks be attached to loads?	5;21
22	Will sanding/cutting/grinding of composites occur?	22

23	Will resins or solvents be used, mixed, cleaned, or stored?	23
24	Will portable generators be used?	24
25	Will curing ovens or hot boxes be used?	25

Example Screens